

# Introducción al método de regresión para Investigación Urbana

Métodos cuantitativos aplicados a estudios urbanos II - MEU  
UTDT

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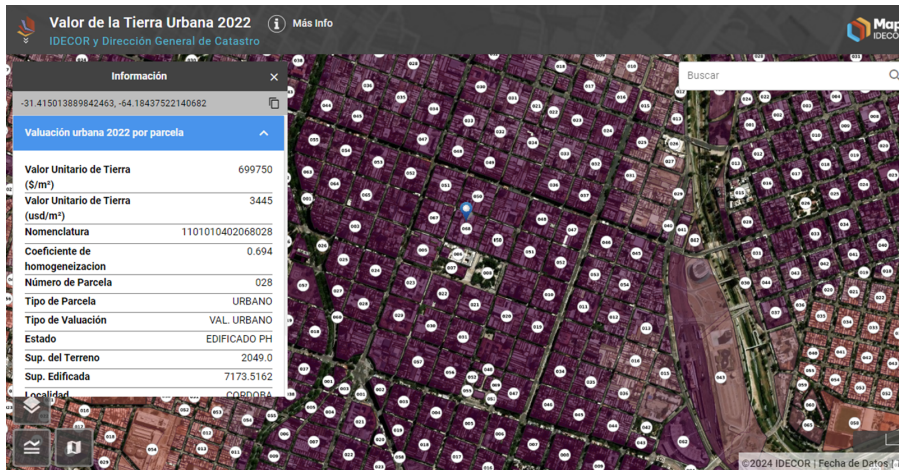
June 17, 2026

# Usos de los Modelos de Regresión

1. Predicción
2. Extrapolación (una forma de predicción)
3. Inferencia causal
  - ▶ Experimental
  - ▶ Cuasi-experimental y observacional

# Predicción

## Valuación de terrenos



<https://mapascordoba.gob.ar/viewer/mapa/401>

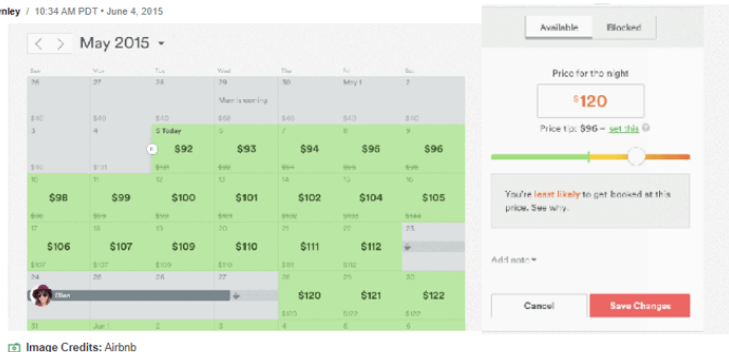
# Predicción

## Recomendación de valor de alquiler

Startups

## Airbnb Adds A Pricing Recommendation Tool For Renters

Matthew Lynley / 10:34 AM PDT • June 4, 2015



<https://techcrunch.com/2015/06/04/airbnb-adds-a-pricing-recommendation-tool-for-renters/>

# Extrapolación

## Extrapolación entre subgrupos

Forecasting elections with non-representative polls

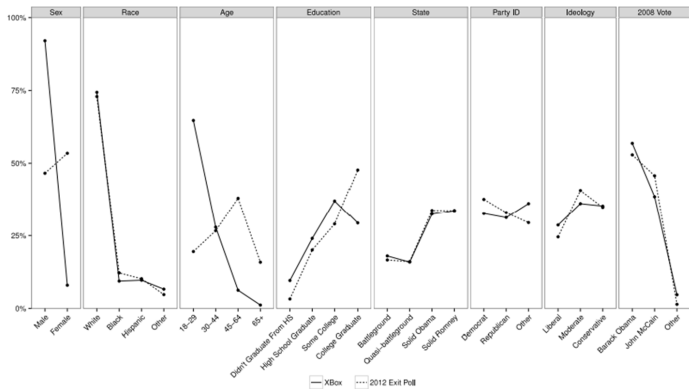
Wei Wang<sup>a,\*</sup>, David Rothschild<sup>b</sup>, Sharad Goel<sup>b</sup>, Andrew Gelman<sup>a,c</sup>

### A B S T R A C T

Election forecasts have traditionally been based on representative polls, in which randomly sampled individuals are asked who they intend to vote for. While representative polling has historically proven to be quite effective, it comes at considerable costs of time and money. Moreover, as response rates have declined over the past several decades, the statistical benefits of representative sampling have diminished. In this paper, we show that, with proper statistical adjustment, non-representative polls can be used to generate accurate election forecasts, and that this can often be achieved faster and at a lesser expense than traditional survey methods. We demonstrate this approach by creating forecasts from a novel and highly non-representative survey dataset: a series of daily voter intention polls for the 2012 presidential election conducted on the Xbox gaming platform. After adjusting the Xbox responses via multilevel regression and poststratification, we obtain estimates which are in line with the forecasts from leading poll analysts, which were based on aggregating hundreds of traditional polls conducted during the election cycle. We conclude by arguing that non-representative polling shows promise not only for election forecasting, but also for measuring public opinion on a broad range of social, economic and cultural issues.

# Extrapolación

## Extrapolación entre subgrupos

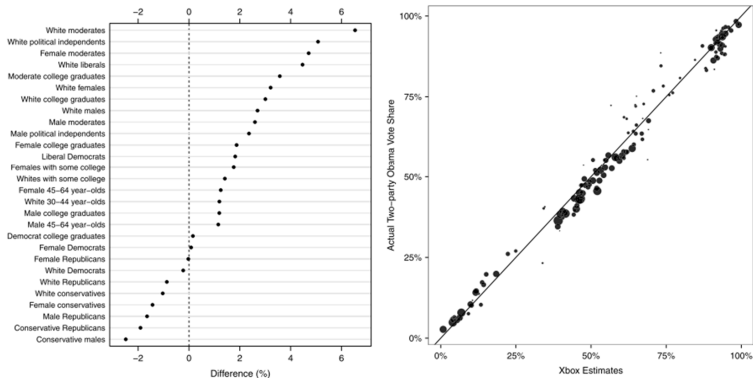


**Fig. 1.** A comparison of the demographic, partisan, and 2008 vote distributions in the Xbox dataset and the 2012 electorate (as measured by adjusted exit polls). As one might expect, the sex and age distributions exhibit considerable differences.

# Extrapolación

## Extrapolación entre subgrupos

**Fig. 5.** Comparison of the two-party Obama vote share for various demographic subgroups, as estimated from the 2012 national exit poll and from the Xbox data on the day before the election.



# Inferencia Causal

## Análisis de experimentos



Journal of Urban Economics  
Volume 142, July 2024, 103412



### JUE insight: The unintended effect of Argentina's subsidized homeownership lottery program on intimate partner violence

Bruno Cardinale Lacomarsino<sup>a</sup>, Martin A. Rossi<sup>b</sup>  

#### Abstract

We study a natural experiment in Argentina, where low-income women were selected through a lottery system to receive a house and a heavily subsidized long-term mortgage. We exploit the random assignment to estimate the causal link between subsidized homeownership programs and intimate partner violence (IPV). Our analysis utilizes administrative records of the population of women applicants to assess the impact of homeownership on IPV, differentiating between women under joint-ownership contracts with their partners and those under single-ownership contracts. We find that the program causes an increase in IPV for women under joint-ownership contracts and a decrease in IPV for women under single-ownership contracts. Our results highlight the importance of considering the design of subsidized homeownership programs and explicitly incorporating measures to facilitate exit from conflicting relationships.

https:

[//www.sciencedirect.com/science/article/abs/pii/S0094119023000827](https://www.sciencedirect.com/science/article/abs/pii/S0094119023000827)

# Inferencia Causal

## Análisis de experimentos

|                    | Dependent variable: IPV |                     |                   |                    |                  |                  |                      |                      |
|--------------------|-------------------------|---------------------|-------------------|--------------------|------------------|------------------|----------------------|----------------------|
|                    | Joint ownership         |                     | Women ownership   |                    | Joint ownership  |                  | Women ownership      |                      |
|                    | (1)                     | (2)                 | (3)               | (4)                | (5)              | (6)              | (7)                  | (8)                  |
| Beneficiary        | 0.173***<br>(0.049)     | 0.196***<br>(0.047) | -0.113<br>(0.070) | -0.116*<br>(0.069) | 0.072<br>(0.122) | 0.069<br>(0.127) | -0.608***<br>(0.172) | -0.501***<br>(0.146) |
| Control group mean | 0.114                   | 0.114               | 0.239             | 0.239              | 0.386            | 0.386            | 0.887                | 0.887                |
| Control variables  | No                      | Yes                 | No                | Yes                | No               | Yes              | No                   | Yes                  |
| Sample             | Admin                   | Admin               | Admin             | Admin              | Survey           | Survey           | Survey               | Survey               |
| Observations       | 290                     | 290                 | 147               | 147                | 85               | 85               | 37                   | 37                   |

*Notes:* Heteroskedastic-robust standard errors in parentheses. The 437 observations correspond to the sample of eligible non-attritor women. Dependent variable equals 1 if the women reported IPV in the administrative data and zero otherwise in columns 1 to 4. Dependent variable is equal to 1 if the woman reported IPV in the survey and zero otherwise in columns 5 to 8. The main independent variable equals 1 if the woman is a beneficiary of the program and zero otherwise. The instrument equals 1 if the woman was assigned by the lottery as a beneficiary of the program and zero otherwise. The set of controls is the same as in Table 3. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

# Inferencia Causal

## Análisis de experimentos

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# Representación del modelo de renovación como una función

$$\text{renovación}_i = f(\text{conectividad}_i, \text{nivel socioeconómico}_i, \text{intangibles}_i, \text{equipamientos}_i)$$

- ▶  $i$  (la unidad de análisis) refiere a un radio censal.
- ▶  $f()$  es una función cuya forma desconocemos.
- ▶ *renovación* se aproximará con el % de parcelas renovadas (cambiaron uso, altura, etc.)
- ▶ *conectividad* se aproximará con distancia a estaciones de subte/tren etc.
- ▶ *nivel socioeconómico* se aproximará por alguna variable del censo (% población universitaria, % NBI, etc)
- ▶ *equipamientos*, aproximada, por ejemplo, con la distancia promedio a un equipamiento (escuelas, centros de salud, etc.)

# Representación del modelo de renovación como una función lineal

$$\begin{aligned}\text{porcentaje\_parcelas\_renovadas}_i = & \beta_0 + \beta_1 \text{distancia\_transporte}_i \\ & + \beta_2 \text{NBI}_i + \beta_3 \text{clima\_barrial}_i \\ & + \beta_4 \text{distancia\_equipamientos}_i + \varepsilon_i\end{aligned}$$

- ▶ Proponemos una forma lineal para la función en cuestión que podremos estimar.
  - ▶ El supuesto de linealidad es restrictivo, pero veremos que admite igualmente relaciones no lineales.
- ▶ Cada coeficiente  $\beta_j$  representa la *contribución* de la variable  $X_j$  a la variable dependiente  $Y$ .
- ▶ Estos coeficientes podrán ser estimados a partir de la muestra de datos.
- ▶  $\varepsilon_i$  es el error aleatorio de la observación  $i$ .

# Base de datos necesaria para la estimación

- ▶ En este caso, la base de datos tendría una observación por cada radio censal.
- ▶ Se corresponde con una base de tipo "corte transversal", esto es, una observación es una instancia de la unidad de análisis.

| id_radio_censal | porcentaje_parcelas_renovadas | distancia_transporte | NBI      | clima_barrial | distancia Equipamientos |
|-----------------|-------------------------------|----------------------|----------|---------------|-------------------------|
| 1               | 0.911746                      | 8.069655             | 0.892558 | 0.695538      | 1.995177                |
| 2               | 0.564947                      | 4.596649             | 0.419448 | 0.849102      | 4.199008                |
| 3               | 0.710200                      | 5.465771             | 0.780366 | -0.293967     | 0.942703                |
| 4               | 0.382642                      | 4.328160             | 0.476336 | -0.071599     | 3.362302                |
| 5               | 0.200962                      | 0.438999             | 0.497540 | -1.517874     | 4.885035                |

## Ejemplo: Modelización hedónica de precios de terrenos

- La modelización hedónica parte de la idea teórica que el valor de un terreno o inmueble es una función de sus características observables.

$$\text{precio}_i = f(\text{superficie}_i, \text{regulación}_i, \text{distanciaCDN}_i, \text{Amenidades}_i)$$

$$\begin{aligned} \text{precio}_i = & \beta_0 + \beta_1 \text{superficie}_i + \beta_2 \text{regulación}_i \\ & + \beta_3 \text{distanciaCDN}_i + \beta_4 \text{Amenidades}_i + \varepsilon_i \end{aligned}$$

# Ejemplo modelo hedónico

## Base de datos

| FECHA      | CALLE     | NUMERO | M2   | DOLARES  | U_S_M2 | FOT | COTIZ | BARRIOS      | COMUNA   | DIRECCION   | CODIGO_P | CODIGO_POSTAL_ARGENTINO |
|------------|-----------|--------|------|----------|--------|-----|-------|--------------|----------|-------------|----------|-------------------------|
| 13/06/2017 | 15 DE NOV | 1600   | 330  | 340000   | 1030,3 | 3   | 17,5  | CONSTITUCION | COMUNA 1 | 15 DE NOV   | 1130     | C1130ABJ                |
| 8/6/2017   | ANTARTIDA | 1400   | 6885 | 35000000 | 5083,5 | 0   | 17,5  | RETIRO       | COMUNA 1 | ANTARTIDA   | 1104     | C1104ACN                |
| 9/6/2017   | BELGRANCO | 800    | 1720 | 7500000  | 4360,5 | 0   | 17,5  | MONSERRATE   | COMUNA 1 | BELGRANCO   | 1092     | C1092AAU                |
| 12/6/2017  | BRASIL    | 1500   | 800  | 600000   | 750    | 0   | 17,5  | CONSTITUCION | COMUNA 1 | BRASIL 1500 | 1154     | C1154AAZ                |
| 7/6/2017   | CALVO, CA | 614    | 254  | 800000   | 3149,6 | 0   | 17,5  | SAN TELMO    | COMUNA 1 | CALVO, CA   | 1102     | C1102AAN                |
| 12/6/2017  | CALVO, CA | 900    | 135  | 280000   | 2074,1 | 5   | 17,5  | CONSTITUCION | COMUNA 1 | CALVO, CA   | 1102     | C1102AAR                |
| 12/6/2017  | CALVO, CA | 1100   | 458  | 650000   | 1419,2 | 5   | 17,5  | CONSTITUCION | COMUNA 1 | CALVO, CA   | 1102     | C1102AAV                |
| 12/6/2017  | CALVO, CA | 1500   | 1356 | 2200000  | 1622,4 | 0   | 17,5  | CONSTITUCION | COMUNA 1 | CALVO, CA   | 1102     | C1102ABD                |
| 9/6/2017   | CALVO, CA | 1600   | 275  | 390000   | 1418,2 | 0   | 17,5  | CONSTITUCION | COMUNA 1 | CALVO, CA   | 1102     | C1102ABF                |
| 12/6/2017  | CALVO, CA | 1600   | 520  | 460000   | 884,6  | 3,4 | 17,5  | CONSTITUCION | COMUNA 1 | CALVO, CA   | 1102     | C1102ABF                |
| 12/6/2017  | CASEROS / | 1701   | 1200 | 2700000  | 2250   | 0   | 17,5  | CONSTITUCION | COMUNA 1 | CASEROS /   | 1152     | C1152ABA                |
| 12/6/2017  | CEVALLOS  | 200    | 620  | 450000   | 725,8  | 4   | 17,5  | MONSERRATE   | COMUNA 1 | CEVALLOS    | 1077     | C1077AAD                |
| 8/6/2017   | CEVALLOS  | 300    | 1400 | 2100000  | 1500   | 0   | 17,5  | MONSERRATE   | COMUNA 1 | CEVALLOS    | 1077     | C1077AAF                |
| 9/6/2017   | CEVALLOS  | 345    | 1400 | 2100000  | 1500   | 0   | 17,5  | MONSERRATE   | COMUNA 1 | CEVALLOS    | 1077     | C1077AAG                |

Figure: Base de datos de terrenos

- ▶ En este caso, la base de datos tendría una observación por cada radio censal.
- ▶ Se corresponde con una base de tipo "corte transversal", esto es, una observación es una instancia de la unidad de análisis.

# Notación para modelización usando regresión

$$\underbrace{Y_i}_{\substack{\text{Variable} \\ \text{a} \\ \text{explicar} \\ \text{o} \\ \text{Dependiente}}} = \underbrace{\beta_0}_{\substack{\text{Constante} \\ \text{o inter-} \\ \text{cepto}}} + \beta_1 \cdot \underbrace{X_{1,i}}_{\substack{\text{Variable} \\ \text{explicativa} \\ 1 \\ \text{(independiente)}}} + \beta_2 \cdot \underbrace{X_{2,i}}_{\substack{\text{Variable} \\ \text{explicativa} \\ 2 \\ \text{(independiente)}}} + \cdots + \underbrace{\varepsilon_i}_{\text{Error}}$$

- ▶  $\beta_0, \beta_2, \dots, \beta_k$  son los coeficientes a ser estimados.
- ▶ Una vez estimados los denotamos con  $\hat{\beta}_0, \hat{\beta}_2, \dots, \hat{\beta}_k$ .
- ▶ Los coeficientes medirán la contribución a la variable explicada por cada unidad de la variable explicativa.

# Explicación y Predicción

## ► Explicación

- Los coeficientes estimados  $\hat{\beta}_0, \hat{\beta}_2, \dots, \hat{\beta}_k$  son la base para la explicación.
- Si bien estrictamente hablando, los coeficientes solo capturan variaciones, y por lo tanto no representan necesariamente causalidad, son la base bajo las cuales buscaremos identificar efectos causales.

## ► Predicción

- La predicción se obtiene reemplazando los coeficientes estimados en la ecuación, y utilizando estos componentes para calcular el valor de que predice el modelo.

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_{1,i} + \hat{\beta}_2 X_{2,i} + \dots + \hat{\beta}_k X_{k,i}$$

# Medidas de Bondad de Ajuste

$R^2$

- ▶ El  $R^2$  es una medida de la bondad de ajuste del modelo en su conjunto.
- ▶ Se define como el cociente entre la varianza explicada por el modelo y la varianza total de la variable dependiente.
- ▶ Se interpreta como la proporción de la varianza de la variable dependiente que es explicada por el modelo.

## Bondad de Ajuste: $R^2$

- Puesto que vale que

$$Y_i = \hat{Y}_i + \hat{\varepsilon}_i$$

- También vale que la varianza de  $Y_i$  es la suma de la varianza de la predicción y la varianza de los residuos.
- Esta igualdad nos permite definir una medida de la bondad de ajuste del modelo: el  $R^2$  es la proporción de la varianza de la variable dependiente que es explicada por el modelo.

$$1 = \frac{\text{Var}(\hat{Y}_i)}{\text{Var}(Y_i)} + \frac{\text{Var}(\hat{\varepsilon}_i)}{\text{Var}(Y_i)}$$

$$R^2 = \frac{\text{Var}(\hat{Y}_i)}{\text{Var}(Y_i)} = 1 - \frac{\text{Var}(\hat{\varepsilon}_i)}{\text{Var}(Y_i)}$$

- Notar que es un valor entre 0 y 1.

# Medidas de Bondad de Ajuste

## Error Cuadrático Medio

- ▶ El error cuadrático medio (MSE) es otra medida de la bondad de ajuste del modelo en su conjunto.
- ▶ Es una medida que busca cuantificar la magnitud de los errores.

$$MSE = \frac{\sum \hat{\epsilon}_i^2}{n - (k + 1)}$$

- ▶ El  $-(k + 1)$  en el denominador es una corrección estadística al promedio, dada por el número de coeficientes estimados.
- ▶ La raíz cuadrada del MSE (o RMSE) es una medida que puede compararse en magnitud con la variable dependiente.